

Metaphors

- [Metaphor is a] thing considered as representative of some other (usually abstract) thing
- Metaphors create relationships between things that are not directly equivalent.
- [Metaphor is a] figure of speech in which a name or descriptive word or phrase is transferred to an object or action different from, but analogous to, that to which it is literally applicable.
- This identifies two important qualities of metaphor: difference and analogy.
- The quality of difference is important because if there were no difference involved, the metaphor would just be a statement of identity.
- The essential difference is what distinguishes the two concepts, the analogy draws them together and allows thinking of one in terms of the other. In this way, interesting unions can be formed which allow us to understand concepts in unique and often useful ways.

Metaphors (Contd.)

- *A device for explaining some concept or thing by asserting its similarity to another concept or thing.*
- Additionally:
 - Metaphor is, by definition, inexact.
 - Metaphor is interpreted contextually at many levels, from single individuals to large scale culture, and everything in-between.
 - Metaphor functions within a system of thought, not in isolation.
 - Metaphor may be one of the key methods humans have for understanding new concepts.

Metaphors and Computing

- Computers are not the only devices made more accessible through metaphor.
- Metaphors are used to help coping with the complexities of any new technology.
- The transition from old technology to new is eased by metaphor.
- When new technologies arrive new metaphors are found to apply to them.
- There are different levels at which one can understand computers.
- People who must deal with each of these levels tend to use metaphor in some way to make the complexity more bearable, from chip-designers through to ordinary users.
- **The user-interface, however, is probably the richest source of metaphors in computers.**
- This is because **the average user of a computer is the least likely of any kind of computer-user to have an understanding of how their computer really functions.**
- **In fact, user understanding may be based almost entirely on metaphors.**

User-Interface Metaphor

- Metaphor is a device for explaining one concept by identifying it with another.
- It is safe to say that, with user-interface metaphors, what is being explained is the underlying functionality and characteristics of the software which the interface gives access to.
- *A device for explaining some system functionality or structure by asserting its similarity to another concept or thing already familiar to the user.*
 - **Example 1:** Consider a familiar example of a user-interface metaphor: the “trashcan”. It would perhaps be something like “file deletion is using a trashcan”. Clearly, this is metaphoric because it uses something familiar to the user (a trash-can) to represent an aspect of the system that might otherwise be confusing (file deletion and management). The trash-can explains certain features of file-deletion in the system. You can place unwanted items into the trash-can to get rid of them, and you can get them back if you are in time. Additionally, the trashcan is a perceivable aspect of the system, using the user’s visual sense to convey the metaphor as well as the kinesthetic sense of “putting” items into it.

User-Interface Metaphor (Contd.)

– **Example 2:** Probably the best known user-interface metaphor is the “desktop metaphor”. This metaphor is present in the graphical interfaces of many modern operating systems. It is perceived through a collection of sub metaphors such as the trash-can, documents and the actual “desktop” surface. The desktop metaphor may be more appropriately called the “office metaphor”, in fact, as the collection of sub-metaphors more accurately characterizes an office working environment. The metaphor exists to help the user understand the function of their computer system as the working of an office, particularly as relates to data management.

- The user encounters “files” and “folders” as ways of representing the more complex data structures that the computer really uses. **This raises the interesting point that metaphors can be applied at different levels. They can be more overarching, organization metaphors, such as the desktop; more specific, like the trash-can metaphor; or even quite subtle and unnoticed, such as the traffic light metaphor.**

Problems with Interface Metaphors

- Sometimes they break conventional and cultural rules
 - e.g. recycle bin placed on desktop
- Can overly constrain designers in the way they conceptualize a problem space
- May conflict with principles of good design
- Forces users to understand the system in terms of the metaphor
- Designers can inadvertently use bad existing designs and transfer the bad parts into the new designs
- Limits designers' imagination in coming up with new conceptual models